

Mitochondrial DNA and neurofilament release as Parkinson's phenotypes in midbrain organoids

Sonia Sabate-Soler^{1,2} , Isabel Rosety¹ , Gemma Gomez-Giro¹ , Jenny Ghelfi² , Soraya Hezzaz² , Anne Grünewald² ,
Jens C. Schwamborn^{1,2} , Javier Jarazo^{1,*} 

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Abstract

Introduction: Parkinson's disease (PD) prevalence keeps growing worldwide, with growing numbers and considerable societal and economic concerns. This study aimed to evaluate the use of microglia-containing midbrain organoids as PD patient-specific systems for the detection of extracellular disease biomarkers.

Materials and methods: We used the cell culture supernatant from healthy control and patient-specific organoids to measure Neurofilament light chain (NfL) levels via ELISA and to study mitochondrial DNA phenotypes through digital polymerase chain reaction (PCR). Organoid tissue was used for immunofluorescence staining and quantification of alpha-synuclein and glial fibrillary acidic protein (GFAP) levels.

Results: Organoids generated from the PD patient line released higher levels of NfL and mitochondrial DNA into the medium. Furthermore, immunofluorescence staining of PD patient-specific organoids showed higher levels of phosphorylated alpha-synuclein as well as GFAP.

Conclusions: Alterations in NfL levels and mitochondrial DNA homeostasis can be measured in the organoid supernatant and are particularly suitable as disease biomarkers and for high-throughput screening approaches.

Keywords: *organoids, mitochondria, Parkinson's disease, neurodegeneration*

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1. Introduction

Parkinson's disease (PD) is the second most common neurodegenerative disorder worldwide, affecting around 3% of the worldwide population over the age of 80 [1]. Different factors play a role in the development of PD, including genetics, such as mutations in the *SNCA*, *LRRK2*, and *PINK1* genes, and environmental factors, such as pesticide exposure. Hereditary forms of PD account for 30% of cases, while 70% are idiopathic [1].

Mitochondrial dysfunction is one of the pathological processes described in PD. Impaired mitochondrial function and metabolism have been observed in idiopathic and monogenic PD forms. Mutations in genes such as *PINK1* and *DJ-1* have been linked to mitochondrial dysfunction in PD [2–4].

One of the main pathophysiological hallmarks of PD is the accumulation of aggregated alpha-synuclein in dopaminergic neurons. The relationship between alpha-synuclein and mitochondrial homeostasis has been studied, and evidence of their relation and functionality has been shown [5, 6]. Mitochondrial homeostasis is not only related to alpha-synuclein aggregation but also to neurodegeneration. Studies conducted over the last few years point to a direct relationship between mitochondrial dysfunction, neuroinflammation, and neurodegeneration [7, 8].

Midbrain organoids (MOs) are advanced, induced pluripotent stem cells (iPSC)-derived, 3D cell culture systems that mimic the human midbrain [9]. Co-culturing MOs with iPSC-derived macrophage precursors leads to microglia-containing midbrain organoids (MGL-MOs), which have a heterogeneous cellular composition and show neuronal functionality [10]. MOs recapitulate PD phenotypes and can be used for drug testing [11].

In this article, we use MGL-MOs as a PD model. We observed phenotypes related to neurodegeneration, as well as alpha-synuclein phosphorylation. We analyzed cell culture supernatants from MGL-MOs with a wild-type (WT) background or from individuals carrying a triplication in the *SNCA* gene (3× *SNCA*), observing neurodegeneration and mitochondrial homeostasis phenotypes. Importantly, significant release of Neurofilament light chain (NfL) was detected. Furthermore, mitochondrial DNA copy number and transcription alterations were detected in those supernatants, opening the door to biomarker detection in cell culture media from human-personalized organoid and assembloid models.

¹OrganoTherapeutics SARL, Esch-Alzette, Luxembourg.

²Luxembourg Centre for Systems Biomedicine, University of Luxembourg, Esch-Alzette, Luxembourg.

*email: javier.jarazo@organo-therapeutics.com

2. Materials and methods

2.1. Microglia-containing midbrain organoids (MGL-MO) generation

Human MOs were generated as described in [12, 13]. In short, 9000 neuro-epithelial stem cells (NESCs) were seeded per well in ultra-low-attachment 96-well plates (FaCellitate, Mannheim, Germany, F202003). They were cultured for 2 days in maintenance medium (N2B27 medium supplemented with 150 μ M Ascorbic acid (Sigma-Aldrich, St. Louis, MO, USA, A4544), 3 μ M CHIR 99021 (Axon Medchem, Groningen, The Netherlands, 1386), 3 μ M Purmorphamine (PMA, Enzo Lifesciences, Farmingdale, NY, USA, ALX-420-045-M001). After that, the medium was switched to neuronal differentiation-inducing medium (N2B27 medium supplemented with 200 μ M Ascorbic acid (Sigma-Aldrich, A4544), 10 ng/ml Brain Derived Neurotrophic Factor (BDNF, Peprotech, Cranbury, NJ, USA, 450-02), 10 ng/ml Glial-Derived Neurotrophic Factor (GDNF, Peprotech, 17814463), 1 ng/ml Transforming growth factor beta-3 (TGF- β 3, Peprotech, 100-36E), 500 μ M db cAMP (Biosynth, St. Gallen, Switzerland, D07996), 1 μ M Purmorphamine (PMA, Enzo Lifesciences, ALX-420-045-M001). The organoids were cultured with neuronal differentiation-inducing medium for 5 days. The day after, MOs were co-cultured with macrophage precursor cells following [10], with some modifications. Soon after, 50,000 macrophage precursor cells were added to each developing organoid, in co-culture medium: Advanced DMEM/F12 (Gibco, Grand Island, NY, USA, 12634028), 1 $\times$ N2 supplement (Thermo Fisher, Waltham, MA, USA, 17502001), 1 $\times$ GlutaMAXTM (Thermo Fisher, 35050061), 50 μ M 2-mercaptoethanol (Thermo Fisher, 11528926), 100 U/ml Penicillin–Streptomycin (Gibco, 15140163), 100 ng/ml IL-34 (Peprotech, 200-34), 10 ng/ml Granulocyte-macrophage colony-stimulating factor (GM-CSF, Peprotech, 300-03), 10 ng/ml BDNF, 10 ng/ml GDNF. MGL-MOs were cultured with co-culture medium for an additional 53 days.

2.2. Media collection and storage

MGL-MO culture medium was collected in 96 Well Polypropylene Sample Storage Microplates (Thermo Scientific, 267334), and sealed with aluminum plate seals (Thermo Scientific, 232698). After that, the collected samples were snap-frozen and stored at -80°C before analysis.

2.3. Mitochondrial DNA analysis

2.3.1. DNA extraction

The MGL-MO medium was collected and snap-frozen for storage. A volume of 90 μ l of culture medium was collected per organoid, and before extraction, samples were diluted by adding 110 μ l of 1 $\times$ PBS (Gibco, 14040-091). A volume of 20 μ l Proteinase K (Qiagen, Hilden, Germany, RP103B) at a concentration of 40 mAU/mg or 600 mAU/ml was added to each sample, and samples were processed following the manufacturer's instructions for the QIAamp DNA Mini Kit (Qiagen, 51304).

2.3.2. Digital PCR

Digital PCR (dPCR) was performed using QuantStudioTM Absolute QTM Digital PCR System (Applied Biosystems (Foster City, CA,

USA), A52864). The primers used for mitochondrial *ND1*, *ND4*, D-loop, and *B2M* are described in [14]. The following probes were used: ND1-VIC (Assay ID: APAANDG, Cat. No. 4448510) targeting mitochondrial ND1; B2M-FAM (Assay ID: APCFGXE, Cat. No. 4332079) targeting nuclear B2M; ND4-ABY/QSY (custom, Article No. CCU002NR, Thermo Fisher) targeting mitochondrial ND4; and DLOOP-JUN/QSY (custom, CCU002NR, Thermo Fisher) targeting the mitochondrial D-loop region.

Samples were prepared by adding 2 μ l of the 5 $\times$ Absolute QTM DNA Digital PCR Master Mix (Applied Biosystems, A52490), 0.166 μ l of the corresponding 60 $\times$ primer probes, and MilliQ water to 5 μ l of DNA, in a final volume of 10 μ l per reaction. The triplex qPCR program was run, with an initial 10 min run at 96°C followed by 40 cycles of 5 s at 96°C and 30 s at 60°C .

2.4. Neurofilament light chain (NfL) ELISA

MGL-MO culture medium was used to perform NfL ELISA. The assay was performed using the NF-light[®] kit (Quanterix, 10-7001), following the manufacturer's instructions. The concentrations of NfL were calculated by plotting the average optical density (λ 450– λ reference) against the standard curve concentrations.

2.5. Immunofluorescence staining of MGL-MO sections

MGL-MOs were fixed in 4% formaldehyde (Sigma-Aldrich, 1004965000) overnight at 4°C . They were then embedded in 3% low-melting-point agarose (Thermo Scientific, R0801) and sectioned into 60 μ m sections using a Vibratome (Leica (Wetzlar, Germany), VT1000 S).

For immunofluorescence staining, MGL-MO sections were incubated for 1 h in blocking and permeabilization solution (0.5% Triton (Sigma-Aldrich, 93443), 5% goat serum (Invitrogen, Carlsbad, CA, USA, 10000C) in 1 $\times$ PBS) at room temperature. After that, they were incubated for 48 h with antibodies (Table S1) in primary antibody solution (0.2% Triton, 5% goat serum, 0.1% Na-Azide (Morphisto, Offenbach, Germany, 18189.00100) in 1 $\times$ PBS) at 4°C . Sections were washed three times for 20 min in 0.1% Tween (Sigma-Aldrich, P7949) in PBS at room temperature. Then, they were incubated for 1 h with secondary antibodies (Table S1) in secondary antibody solution (0.1% Tween, 5% goat serum, 0.1% Na-Azide in 1 $\times$ PBS) at room temperature protected from light. Three washes in 0.1% Tween in PBS were done, and sections were rinsed with milliQ H₂O and mounted with Fluoromount-G (Thermo Scientific, 00-4958-020) mounting medium. After drying, sections were imaged using a Yokogawa CellVoyager CV8000 High-Content Screening System (Yokogawa Electric Corporation, Tokyo, Japan).

3. Results

3.1. Patient-derived MGL-MOs from 3 $\times$ SNCA patients show increased neuronal death

MOs were generated from human iPSCs, as described previously [12, 13]. A wild-type and 3 $\times$ SNCA iPSC lines (Table S2) were used for the generation of macrophage and neural precursor cells. The latter were used for the generation of MOs, which were then co-cultured with their line-matched macrophage precursor cells for MGL-MO generation (Figure 1A). Therefore, organoids from one wild-type and one 3 $\times$ SNCA line were generated for this study.

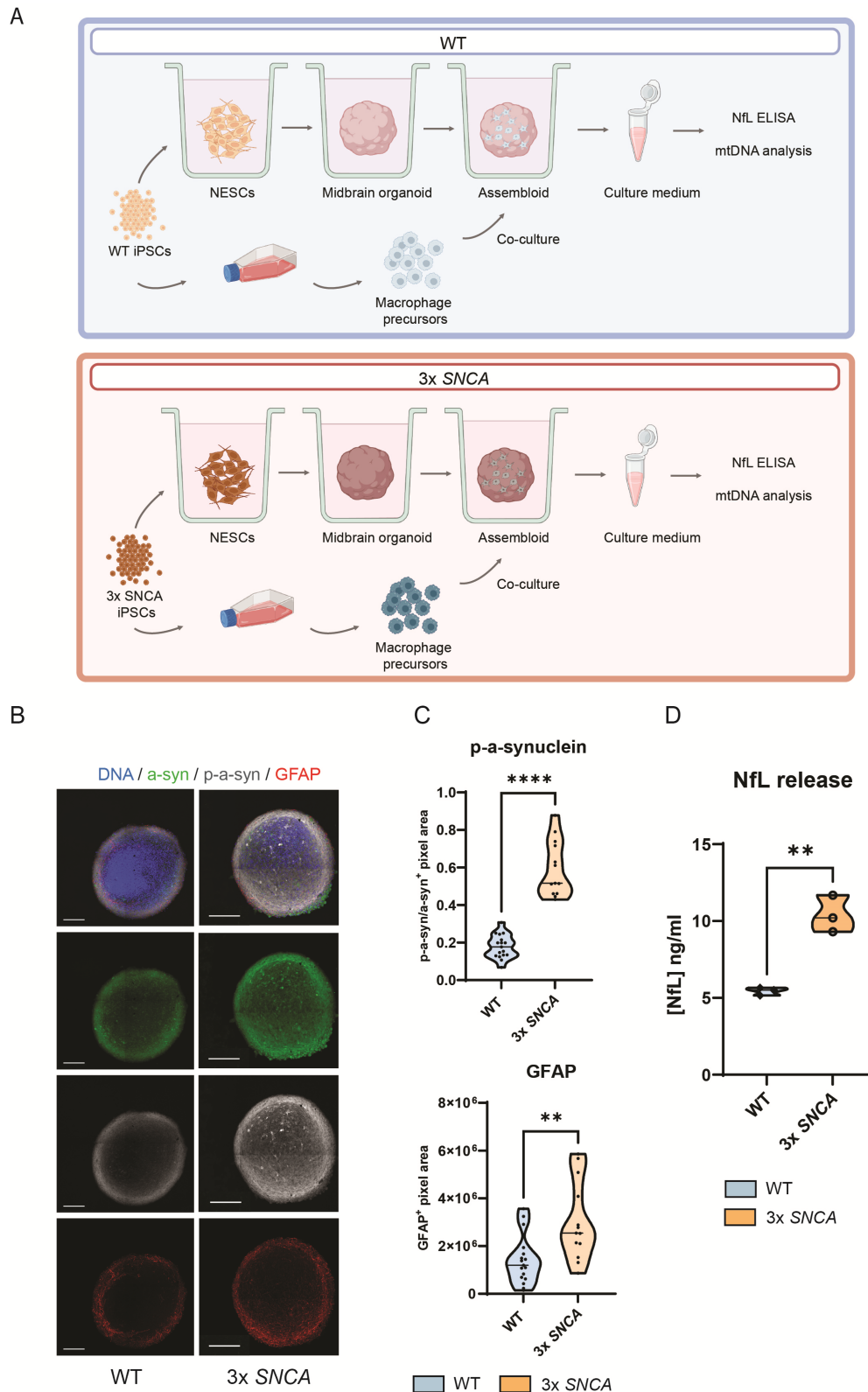


Figure 1 • Human microglia-containing midbrain organoids (MGL-MOs) from the 3× *SNCA* group show neurodegeneration and synucleinopathy-related phenotypes. (A) Diagram showing the culture and analysis methods for the wild-type (WT) group (**upper panel**) and 3× *SNCA* (**bottom panel**). (B) Immunofluorescence images showing WT (**left panels**) and 3× *SNCA* (**right panels**) sections stained for alpha-synuclein (a-syn, green), phosphorylated alpha-synuclein (p-a-syn, grey), and glial fibrillary acidic protein (GFAP) (red). Scale bar = 200 μm. (C) Violin plots showing the image analysis quantification results for phosphorylated alpha-synuclein (p-a-synuclein, **upper graph**) and GFAP (**bottom graph**). Each dot represents an organoid; $n = 13-17$. (D) Violin plot showing released neurofilament light chain (NfL) in culture supernatants, measured by ELISA. Data shown are from MGL-MOs at day 60, with one WT and one 3× *SNCA* represented. Violin plots show median and quartiles. Each dot represents a cell culture batch; $n = 3$. ** = $p \leq 0.01$; **** = $p \leq 0.0001$. mtDNA: mitochondrial deoxyribonucleic acid, NESCs: neuro-epithelial stem cells, iPSCs: induced pluripotent stem cells. Panel A created in BioRender. Sabate Soler, S. (2025) <https://BioRender.com/n25c744>.

The presence of astrocytes and microglia in the system is of high value for assessing PD phenotypes. Astrocyte senescence has been linked to neurodegeneration in PD [15], and the regulation of astrocytic activity has led to the rescue of PD symptoms in mice [16]. Microglia participate in many brain processes, including the control of neuronal activity [17]. Microglia in PD have been associated with dopaminergic-specific neurodegeneration [18], and interactions between microglia and astrocytes have been reported as contributors to neurodegeneration [19, 20]. Because of that, we believe that using a complete midbrain organoid model with astrocytes and microglia (MGL-MOs) could lead to better observation of PD phenotypes.

To validate PD-related phenotypes, we performed immunofluorescence staining of MGL-MO sections from day 60 of culture. We used a-syn (alpha-synuclein), p-a-syn (alpha-synuclein phosphorylated at Serine 129), and glial fibrillary acidic protein (GFAP) antibodies (**Figure 1B**) and used a published image analysis pipeline to quantify the positive marker area [21]. We observed significantly higher levels of the phosphorylated a-syn proportion to total synuclein in the $3\times$ SNCA MGL-MO. This indicates possible alpha-synuclein aggregation in the model, characteristic of PD and other synucleinopathies [22–24]. Furthermore, the GFAP⁺ area was significantly higher in the $3\times$ SNCA group (**Figure 1C**). Increased GFAP⁺ astrocyte levels have been previously observed in PD, both in mice and human brain tissue [25–27]. The higher GFAP⁺ area correlates with previous studies on $3\times$ SNCA human organoids [28], and could indicate an astrogliosis process in the PD group.

MGL-MOs were cultured for 60 days, after which culture supernatant was collected for NfL release analysis. We observed a significant increase in NfL release in the $3\times$ SNCA group, indicating neurodegeneration (**Figure 1D**). NfL release has been established as a biomarker for neurodegeneration and phenotype severity in CSF and plasma samples from PD patients [29–32]. This parameter has been detected in cell culture supernatant samples from human cortical organoids [33]. However, it has never

been observed in patient-specific midbrain organoid cultures. For the first time, we report a significant increase in released NfL in MGL-MOs, demonstrating the clinical translatability of the model and opening a door to PD biomarker detection in iPSC-derived MGL-MOs.

3.2. Patient-derived MGL-MOs release higher levels of mitochondrial DNA

At day 60 of culture, MGL-MO medium was collected for mitochondrial DNA extraction and digital PCR (dPCR) analysis. Three parameters were assessed: mitochondrial DNA copy number, 7S DNA-mediated transcription, and the occurrence of major arc deletions. Via digital PCR, the gene abundance of *ND1* (NADH dehydrogenase 1), a mitochondrial gene, and *B2M* (Beta-2 microglobulin), a nuclear gene, were assessed. The mitochondrial DNA copy number was calculated by determining the *ND1/B2M* ratio. The copy number levels detected in culture supernatants from the $3\times$ SNCA group were statistically higher than those from the WT (**Figure 2A**). When normalizing the *ND1* abundance to the organoid size, we observed a tendency to an increased copy number in the $3\times$ SNCA group (**Figure 2A**), following the same trend as the *ND1/B2M* ratio. Since the data normalization to *B2M* in the medium takes into account DNA release, coming most likely from cell death, the *ND1/B2M* ratio gives valuable information on the mitochondrial DNA abundance due to active release rather than cell death. The fact that the differences in *ND1/B2M* are significant and the *ND1/size* are not could highlight a process of active mitochondrial DNA release in the $3\times$ SNCA group. Overall, the findings on the copy number in the media correlate with previous studies that report alterations in mitochondrial homeostasis in PD [34–36]. Increased blood levels of mitochondrial DNA have been detected in PD patients [34]. Furthermore, higher levels of mitochondrial DNA have been found in genetic PD patients compared to idiopathic forms, leading to a more accentuated neuroinflammatory phenotype [35].

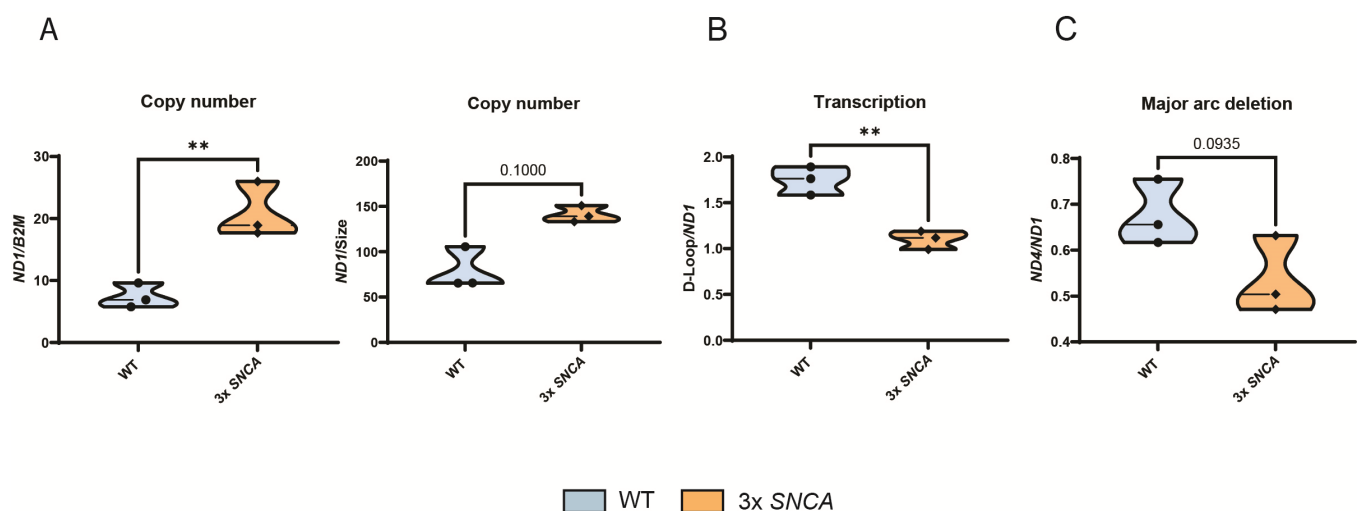


Figure 2 • Microglia-containing midbrain organoids (MGL-MOs) from the $3\times$ SNCA group show alterations in mitochondrial DNA dynamics at day 60. (A) Copy number measured by *ND1/B2M* ratio, showing a significant increase in $3\times$ SNCA supernatant and copy number measured by *ND1/organoid size*, showing a tendency to an increased ratio in the $3\times$ SNCA group. (B) Transcription rate measured by D-loop/*ND1*, showing lower levels in $3\times$ SNCA organoid supernatant. (C) Major arc deletion ratio measured by *ND4/ND1*, showing a tendency for a decrease in $3\times$ SNCA supernatant. Data shown is from MGL-MOs at day 60, with one wild-type (WT) and one $3\times$ SNCA represented. Violin plots show median and quartiles. Each dot represents a cell culture batch; n = 3. ** = p ≤ 0.01. *ND1*: NADH dehydrogenase 1, *B2M*: beta-2 microglobulin, *ND4*: NADH dehydrogenase subunit 4, D-loop: displacement loop.

Mitochondrial transcription was assessed by quantifying the abundance of 7S DNA in the D-loop (displacement loop) region relative to *ND1*. 7S DNA is a fragment that serves as a natural “primer” during mtDNA transcription [37]. When present, the D-loop assumes a triple-strand state that can be detected by dPCR. Our results showed a significantly lower transcription rate in the supernatant from the 3× *SNCA* group compared to the WT (**Figure 2B**). Finally, the major arc deletion rate was assessed by dividing *ND4* (NADH dehydrogenase 4) by *ND1*. Since the *ND4* gene is located in the major arc of the mitochondrial DNA genome, which is commonly subjected to deletions, the *ND4/ND1* ratio indicates the deletion rate in mitochondrial DNA. There was no significant difference for this parameter, although a tendency to decrease was observed in the 3× *SNCA* group, indicating a possible higher deletion number in the 3× *SNCA* group (**Figure 2C**). A significant decrease in mitochondrial transcription (via transcription factor A mitochondrial, TFAM, expression) as well as multiple deletions [36, 38] have been previously reported in neuronal cultures from post-mortem human PD tissue.

These results illustrate the applicability of MGL-MOs in extracellular biomarker detection. Future studies with larger datasets could explore potential correlations between neurodegeneration markers and mitochondrial alterations in this system. A limitation of the present study is that the control iPSC line used for comparison is not isogenic to the patient-derived line. Consequently, genetic background differences between lines may contribute to some of the observed phenotypic variability. While the use of multiple independent lines helps mitigate line-specific effects, future studies employing isogenic control pairs generated through genome editing would allow for a more precise dissection of mutation-specific phenotypes.

Overall, this study shows how MGL-MOs are a successful tool for translating clinical phenotypes into in vitro research. Furthermore, the fact that the observed phenotypes were detected in cell culture media, with uncomplicated manipulation and short experimental time, indicates that these parameters are not only translatable but also suitable for high-throughput analysis. Finally, these findings have implications for the potential application of NFL and released mitochondrial DNA as biomarkers.

4. Conclusions

Human MGL-MOs are an effective model for detecting neurodegeneration via NFL abundance in cell culture supernatants. Furthermore, alterations in mitochondrial copy number and transcription were detected in supernatants from a 3× *SNCA* PD patient, which correlates with studies where human PD blood samples were analyzed. In summary, our results indicate that organoids could be candidate models for detecting extracellular PD biomarkers in vitro with straightforward techniques compatible with high-throughput analyses.

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Author contributions

Conceptualization, S.S.-S., J.C.S. and J.J.; methodology, S.S.-S., I.R., G.G.-G., J.G. and S.H.; validation, S.S.-S., I.R., G.G.-G., J.G. and S.H.; formal analysis, S.S.-S. and J.J.; investigation, S.S.-S., I.R., G.G.-G., J.G. and S.H.; resources, A.G., J.C.S. and J.J.; data curation, S.S.-S. and J.J.; writing—original draft preparation, S.S.-S.; writing—review and editing, S.S.-S., A.G., J.C.S. and J.J.; visualization, S.S.-S.; supervision, A.G., J.C.S. and J.J.; project administration, S.S.-S. and J.J.; funding acquisition, A.G., J.C.S. and J.J. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

J.C.S. and J.J. are co-founders of OrganoTherapeutics SARL. The funding sponsors played no role in the study’s conception, execution, or interpretation, or in the preparation of this article.

Data availability statement

The data supporting the findings of this publication can be made available upon request.

Institutional review board statement

The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Institutional Review Board Luxembourg’s Comité National D’Ethique De Recherche (CNER) (protocol code: 202406/04; date of approval: 19 December 2024).

Supplementary materials

The supplementary materials are available at <https://doi.org/10.20935/AcadNeurosci8295>. Reference [39] is cited in the Supplementary materials.

Additional information

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References

- Ben-Shlomo Y, Darweesh S, Llibre-Guerra J, Marras C, San Luciano M, Tanner C. The epidemiology of Parkinson's disease. *Lancet*. 2024; 403 (10423): 283–92. doi: 10.1016/S0140-6736(23)01419-8
- Watanabe A, Ogiwara S, Saito M, Mishima M, Yamashina M, Ishitani R, et al. The reaction mechanism for glycolysis side product degradation by Parkinson's disease-linked DJ-1. *J Cell Biol*. 2025; 224 (8): e202411078.
- Wang Z, Chan SW, Zhao H, Miu KK, Chan WY. Outlook of PINK1/Parkin signaling in molecular etiology of Parkinson's disease, with insights into Pink1 knockout models. *Zool Res*. 2023; 44 (3): 559–76. doi: 10.24272/j.issn.2095-8137.2022.406
- Brassard R, Arutyunova E, Takyi E, Espinoza-Fonseca LM, Young HS, Touret N, et al. Transmembrane Parkinson's disease mutation of PINK1 leads to altered mitochondrial anchoring. *J Biol Chem*. 2025; 301 (3): 108253. doi: 10.1016/j.jbc.2025.108253
- Hui S, George J, Kapadia M, Chau H, Bariring Z, Earnshaw R, et al. Mitophagy upregulation occurs early in the neurodegenerative process mediated by α -synuclein. *Mol Neurobiol*. 2024; 61 (11): 9032–42. doi: 10.1007/s12035-024-04131-6
- Hou X, Chen TH, Koga S, Bredenberg JM, Faroqi AH, Delcambre M, et al. Alpha-synuclein-associated changes in PINK1-PRKN-mediated mitophagy are disease context dependent. *Brain Pathol*. 2023; 33 (5): e13175. doi: 10.1111/bpa.13175
- Lurette O, Martín-Jiménez R, Khan M, Sheta R, Jean S, Schofield M, et al. Aggregation of alpha-synuclein disrupts mitochondrial metabolism and induces mitophagy via cardiolipin externalization. *Cell Death Dis*. 2023; 14 (11): 729. doi: 10.1038/s41419-023-06251-8
- Angelova PR, Esteras N, Evans J, Kostic M, Melki R, Prehn JHM, et al. α -Synuclein fibrils per se but not α -synuclein seeded aggregation causes mitochondrial dysfunction and cell death in human neurons. *Redox Biol*. 2025; 86: 103817. doi: 10.1016/j.redox.2025.103817
- Zagare A, Barmpla K, Smajic S, Smits LM, Grzyb K, Grünwald A, et al. Midbrain organoids mimic early embryonic neurodevelopment and recapitulate LRRK2-p. Gly2019Ser-associated gene expression. *Am J Hum Genet*. 2022; 109 (2): 311–27. doi: 10.1016/j.ajhg.2021.12.009
- Sabate-Soler S, Nickels SL, Saraiva C, Berger E, Dubonyte U, Barmpla K, et al. Microglia integration into human mid-brain organoids leads to increased neuronal maturation and functionality. *Glia*. 2022; 70 (7): 1267–88. doi: 10.1002/glia.24167
- Jarazo J, Barmpla K, Modamio J, Saraiva C, Sabaté-Soler S, Rosety I, et al. Parkinson's disease phenotypes in patient neuronal cultures and brain organoids improved by 2-hydroxypropyl- β -cyclodextrin treatment. *Mov Disord*. 2022; 37 (1): 80–94. doi: 10.1002/mds.28810
- Nickels SL, Modamio J, Mendes-Pinheiro B, Monzel AS, Betsou F, Schwamborn JC. Reproducible generation of human midbrain organoids for in vitro modeling of Parkinson's disease. *Stem Cell Res*. 2020; 46: 101870. doi: 10.1016/j.scr.2020.101870
- Smits LM, Reinhardt L, Reinhardt P, Glatz M, Monzel AS, Stanslowsky N, et al. Modeling Parkinson's disease in midbrain-like organoids. *npj Parkinsons Dis*. 2019; 5 (1): 5. doi: 10.1038/s41531-019-0078-4
- Rygiel KA, Grady JP, Taylor RW, Tuppen HAL, Turnbull DM. Triplex real-time PCR—an improved method to detect a wide spectrum of mitochondrial DNA deletions in single cells. *Sci Rep*. 2015; 5: 9906. doi: 10.1038/srep09906
- Jiang SY, Tian T, Yao H, Xia XM, Wang C, Cao L, et al. The cGAS-STING-YY1 axis accelerates progression of neurodegeneration in a mouse model of Parkinson's disease via LCN2-dependent astrocyte senescence. *Cell Death Differ*. 2023; 30 (10): 2280–92. doi: 10.1038/s41418-023-01216-y
- Ryu YK, Park HY, Kim JE, Seo HH, Lee CH, Kim KS. Regulating astrocytic activity in the dorsal striatum mitigates L-dopa-induced dyskinesia in Parkinson's disease. *Sci Rep*. 2025; 15 (1): 26635. doi: 10.1038/s41598-025-12104-5
- Badimon A, Strasburger HJ, Ayata P, Chen X, Nair A, Ikegami A, et al. Negative feedback control of neuronal activity by microglia. *Nature*. 2020; 586 (7829): 417–23. doi: 10.1038/s41586-020-2777-8
- Blasco-Agell L, Pons-Espinal M, Testa V, Roch G, Montero-Muñoz J, Fernandez-Carasa I, et al. LRRK2-mutant microglia and neuromelanin synergize to drive dopaminergic neurodegeneration in an iPSC-based Parkinson's disease model. *Commun Biol*. 2025; 8 (1): 1203. doi: 10.1038/s42003-025-08544-4
- Liddel SA, Gattenplan KA, Clarke LE, Bennett FC, Bohlen CJ, Schirmer L, et al. Neurotoxic reactive astrocytes are induced by activated microglia. *Nature*. 2017; 541 (7638): 481–7. doi: 10.1038/nature21029
- Gotoh M, Miyamoto Y, Ikeshima-Kataoka H. Astrocytic neuroimmunological roles interacting with microglial cells in neurodegenerative diseases. *Int J Mol Sci*. 2023; 24 (2): 1599. doi: 10.3390/ijms24021599
- Bolognin S, Fossépré M, Qing X, Jarazo J, Šančar J, Moreno EL, et al. 3D cultures of Parkinson's disease-

- specific dopaminergic neurons for high content phenotyping and drug testing. *Adv Sci.* 2019; 6 (1): 1800927. doi: 10.1002/advs.201800927
22. Ramalingam N, Brontesi L, Jin S, Selkoe DJ, Dettmer U. Dynamic reversibility of α -synuclein serine-129 phosphorylation is impaired in synucleinopathy models. *EMBO Rep.* 2023; 24 (12): e57145. doi: 10.15252/embr.202357145
 23. Han R, Wang Q, Xiong X, Chen X, Tu Z, Li B, et al. Deficiency of parkin causes neurodegeneration and accumulation of pathological α -synuclein in monkey models. *J Clin Invest.* 2024; 134 (20): e179633. doi: 10.1172/JCI179633
 24. Frigerio I, Morella ML, Winkelaar AB, Vorgia DT, Brevé JJ, Boiten WA, et al. Synaptic enrichment of pSer129 alpha-synuclein correlates with dopaminergic denervation in early-stage Parkinson's disease. *Nat Commun.* 2025; 16 (1): 6630. doi: 10.1038/s41467-025-61052-1
 25. Martinez PA, Martinez VE, Rani S, Murrell M, Javors M, Gelfond J, et al. Impaired aldehyde detoxification exacerbates motor deficits in an alpha-synuclein mouse model of Parkinson's disease. *Brain Behav.* 2023; 13 (9): e3150. doi: 10.1002/brb3.3150
 26. Timoshina YA, Pavlova AK, Voronkov DN, Abaimov DA, Latanov AV, Fedorova TN. Assessment of the behavioral and neurochemical characteristics in a mice model of the premotor stage of Parkinson's disease induced by chronic administration of a low dose of MPTP. *Int J Mol Sci.* 2025; 26 (18): 8856. doi: 10.3390/ijms26188856
 27. Thangavel R, Kaur H, Dubova I, Selvakumar GP, Ahmed ME, Raikwar SP, et al. Parkinson's disease dementia patients: expression of Glia maturation factor in the brain. *Int J Mol Sci.* 2024; 25 (2): 1182. doi: 10.3390/ijms25021182
 28. Muwanigwa MN, Modamio-Chamarro J, Antony PMA, Gomez-Giro G, Krüger R, Bolognin S, et al. Alpha-synuclein pathology is associated with astrocyte senescence in a midbrain organoid model of familial Parkinson's disease. *Mol Cell Neurosci.* 2024; 128: 103919. doi: 10.1016/j.mcn.2024.103919
 29. Bäckström D, Linder J, Jakobson Mo S, Riklund K, Zetterberg H, Blennow K, et al. NfL as a biomarker for neurodegeneration and survival in Parkinson disease. *Neurology.* 2020; 95 (7): e827–38. doi: 10.1212/WNL.00000000000010084
 30. Painous C, Pascual-Diaz S, Muñoz-Moreno E, Sánchez V, Pariente JC, Prats-Galino A, et al. Midbrain and pons MRI shape analysis and its clinical and CSF correlates in degenerative parkinsonisms: a pilot study. *Eur Radiol.* 2023; 33 (7): 4540–51. doi: 10.1007/s00330-023-09435-0
 31. Baiardi S, Rossi M, Giannini G, Mammana A, Polischi B, Sambati L, et al. Head-to-head comparison of four cerebrospinal fluid and three plasma neurofilament light chain assays in Parkinsonism. *npj Parkinsons Dis.* 2025; 11 (1): 98. doi: 10.1038/s41531-025-00951-y
 32. Li L, Yang Y, Zhang E, Chen H, Zhao Z. Serum neurofilament light chain predicts autonomic symptoms in early Parkinson's disease. *Neurodegener Dis Manag.* 2025; 15 (6): 263–70. doi: 10.1080/17582024.2025.2554531
 33. De Paola M, Pischutta F, Comolli D, Mariani A, Kelk J, Lisi I, et al. Neural cortical organoids from self-assembling human iPSC as a model to investigate neurotoxicity in brain ischemia. *J Cereb Blood Flow Metab.* 2023; 43 (5): 680–93. doi: 10.1177/0271678X231152023
 34. Müller-Nedebock AC, Meldau S, Lombard C, Abrahams S, van der Westhuizen FH, Bardien S. Increased blood-derived mitochondrial DNA copy number in African ancestry individuals with Parkinson's disease. *Parkinsonism Relat Disord.* 2022; 101: 1–5. doi: 10.1016/j.parkreldis.2022.06.003
 35. Borsche M, König IR, Delcambre S, Petrucci S, Balck A, Brüggemann N, et al. Mitochondrial damage-associated inflammation highlights biomarkers in PRKN/PINK1 parkinsonism. *Brain.* 2020; 143 (10): 3041–51. doi: 10.1093/brain/awaa246
 36. Chen C, Vincent AE, Blain AP, Smith AL, Turnbull DM, Reeve AK. Investigation of mitochondrial biogenesis defects in single substantia nigra neurons using post-mortem human tissues. *Neurobiol Dis.* 2020; 134: 104631. doi: 10.1016/j.nbd.2019.104631
 37. Nicholls TJ, Minczuk M. In D-loop: 40 years of mitochondrial 7S DNA. *Exp Gerontol.* 2014; 56: 175–81. doi: 10.1016/j.exger.2014.03.027
 38. Grünwald A, Rygiel KA, Hepplewhite PD, Morris CM, Picard M, Turnbull DM. Mitochondrial DNA depletion in respiratory chain-deficient Parkinson disease neurons. *Ann Neurol.* 2016; 79 (3): 366–78. doi: 10.1002/ana.24571
 39. Reinhardt P, Schmid B, Burbulla LF, Schöndorf DC, Wagner L, Glatza M, et al. Genetic correction of a LRRK2 mutation in human iPSCs links parkinsonian neurodegeneration to ERK-dependent changes in gene expression. *Cell Stem Cell.* 2013; 12 (3): 354–67. doi: 10.1016/j.stem.2013.01.008